

advantage. This becomes even more apparent, when compared with the landed cost of green ammonia, which is \$800-1,000 per MT for delivery from Middle East to the Far East. Industry players believe that landed cost of green ammonia exported from India to destinations like Japan or South Korea will be in a similar range, if not lower. Further, as technology matures, production costs are expected to decline and provide further impetus towards making India cost-competitive.

WHY THE URGENCY?

Global hydrogen ambitions, driven by stakeholders rushing to comply with regulations, creates a definitive call to action for all prospective producers of GH₂ worldwide. Upon reviewing global hydrogen capacities and commitments to 'Net Zero' by countries and companies, particularly in HTA sectors, it appears that 2030 will be a pivotal year with significant increases in both demand and supply. Case in point is the EU RE Directive, which mandates 42% of industrial hydrogen to be green by 2030. Other significant markets like Japan and South Korea, too, have set targets. From a supply perspective, too, significant GH₂ production capacity is expected to come online globally by 2030. This makes it imperative for India to invest, develop, and scale the ecosystem. While India may not be the first mover, it can establish itself as the leader by creating cost efficiencies, offering stable supply, and cementing partnerships worldwide.

ENABLING INDIA'S AMBITIONS

To leverage these advantages and fulfil its ambitions of becoming a global hydrogen hub, India must also implement several key enablers:

1. Empower the Grid: Typical transmission infrastructure construction timelines can range up to three years or more, while solar and wind projects are usually completed in under two years. As India scales RE capacity, it also needs to invest in building a robust, reliable, and climate resistant transmission network to effectively evacuate RE power.

2. Boost Electrolyser Manufacturing: Currently, India is short of local electrolyser manufactur-

capital subsidies to help reduce the overall cost of production. More states with GH₂ potential need to join the movement to further encourage investment in this space.

5. Develop Cost-effective GH₂ Export Infra at Ports:

The Indian government has identified Kandla, Thoothukudi, and Paradip ports for development as GH₂ hubs for handling and shipping GH₂/derivatives. As part of the Harit Sagar initia-

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ing capacity and requires 60-100 GW to meet the 5 MTPA target set by the government. The industry, hence, needs to step-up with more R&D and manufacturing capacity to bridge the gap.

3. Mandate Demand to Ramp-up Supply: As entrepreneurs seek to establish and scale GH₂ and derivatives capacity, the government can help accelerate final investment decision (FID) in these projects by assuring a minimum demand offtake through mandates.

4. Incentives & Subsidies: The central and some state governments have provided significant incentives for GH₂/derivatives manufacturing and RE by waiving electricity duties, additional and cross-subsidy surcharges, and

tive, the government has issued guidelines for 13 major ports to establish green ammonia bunkering/refueling facilities by 2035, which will be needed as production rises.

The government has taken many initial steps to kick-start the GH₂ economy in India. However, competition from the EU, Middle East, and Australia is fierce. To stay ahead, we are likely to witness continued launch of concerted programmes in India encompassing regulatory policies and incentives, to drive production and cross-sector adoption as well as export of GH₂.

GH₂ offers India a golden opportunity to emerge as a global energy powerhouse—action is crucial. **BT**

Views are personal